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TWO FLAP SEALER (TFS).

Operations Manual.

TWO FLAP SEALER USER INSTRUCTION MANUAL

READ THIS MANUAL CAREFULLY BEFORE OPERATION



STANDARD FEATURES

DESIGNED AND MANUFACTURED TO COMPLY WITH CE MARK REGULATIONS
MANUFACTURED IN FOOD GRADE STAINLESS STEEL AND ALUMINIUM
ISO STANDARD MAINTENANCE FREE PNEUMATIC SYSTEM
SAFETY INTERLOCK SYSTEM ON ALL REMOVABLE GUARDS
HOT MELT GLUE DABBER SYSTEM
THERMOSTAT TEMPERATURE CONTROL

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SECTION 1

A INTRODUCTION / SAFETY

1. A.1 Preface

This instruction manual is directed to personnel who operate and maintain this machine. This manual should not be regarded as a substitute for a thorough instruction in the installation and operation of the machine.

We recommend that your operating personnel participate in the instruction and training programs provided by T. Freemantle's highly skilled service personnel. This service is normally available during installation and commissioning.

This manual is meant to help you during your daily work on the machine and during maintenance which must be carried out regularly.

All personnel involved in the installation, set-up, operation, maintenance and/or repair of this machine must thoroughly read and fully understand this instruction manual.

If you should have any questions, please contact T. Freemantle Ltd. The manual is to be stored in a place near to the machine and must be easily accessible for the operator.

Please operate your machine in accordance with the instructions given in this manual. This will guarantee trouble-free operation and a high availability of your machine.

1. A.2 Safety

a) General Information

This machine has been designed and constructed to meet a high technical standard and is therefore safe and reliable in operation.

Nevertheless the machine can become dangerous

- ◇ if operated improperly by personnel not trained and instructed thoroughly.
- ◇ if the instructions given in this operation and maintenance manual are not strictly observed.

We specifically point out that you should only

- ◇ operate
- ◇ service
- ◇ maintain
- ◇ supervise

the machine if you have been specially trained and instructed to do so.

The machine must only be operated with all safety devices in their proper location and in their operating order. A Safety Interlock System is fitted to all opening guards, inhibiting machine operation if guards are opened.

No modifications must be made to the machine without prior permission of T. Freemantle Ltd.

Note the following when installing the machine:

Hazardous positions between the machine and existing equipment have to be provided with any necessary protection by the customer operating the machine.

Only use products agreed upon with T. Freemantle Ltd. No liability will be accepted for damage arising from improper use and nonobservance of the instructions given in this manual. The responsibility must be assumed by the user.

Hot Melt Glue Dabber Applicators are situated within the guarding of the machine, and are electrically heated. The operating temperatures are factory set according to required glue specification.

CAUTION! IT IS ADVISED THAT ALL USERS READ THE SAFETY INFORMATION CONTAINED IN **SECTION 4A** REGARDING HOT MELT ADHESIVE

WARNING! DO NOT TOUCH ANY PART OF THE HOT MELT UNIT DURING OPERATION

Note! Care should be taken not to contaminate the glue tanks as this may cause gluing issues.

b) Operation

The machine must only be operated by persons

- ◇ who have been properly trained and instructed in their use.
- ◇ who will perform their duty in reliable manner.

It is incumbent upon the company operating the machine to ensure that the instructions given in this manual are strictly followed. Where necessary the company is to issue supplementary instructions.

c) **Maintenance**

Do not perform any work on the machine while it is in operation. If this should be necessary, however, for example during

- ◇ installation
- ◇ maintenance
- ◇ elimination of faults

operate the machine in a safe manner by setting up personal safe operating practices.

Set up a precise working plan if more than one person is to work on the machine at the same time. Appoint a person who will be responsible

- ◇ for coordinating the work to be done
- ◇ that safety instructions are strictly followed

The company operating the machine is responsible for providing an efficient safety program for the machine.

The machine must be operated appropriate to its intended use.

The operating procedures applied must be appropriate to the technical possibilities of the machine.

Safety and protection devices must not be changed or rendered ineffective.

Check the safety devices on the machine

- ◇ at least once a week
- ◇ after rebuilding the machine
- ◇ after major maintenance or repair work
- ◇ after replacing or repairing safety devices

d) **Cleaning**

The machine is designed to be safe to construct, install, operate and maintain, consistent with current legislation. Consideration has been given to the ergonomics of machinery use, i.e. to achieve the most effective safe performance of the machinery and its operator.

The machine is manufactured from only corrosion-resistant (corrosion-resistant coated) materials that can be steam cleaned over and over. All electrical wiring is to I.P. 65, BS 2771 (part .1) and EN60204 (part .1) standard.

Included in this machine manual are instructions for safe starting up, installation, adjustment, maintenance, training and hints for safe and efficient day to day operator input.

T. FREEMANTLE LTD have ensured, so far as is reasonably practicable, that the machinery is so designed and constructed that it will be safe and without risks to health and safety at all times when it is being set, used, cleaned or maintained.

All guarding (fixed or otherwise) should not be removed before or during operation of the machinery. All accessible parts of the machine are guarded (sliding doors or hinged guards) and are electronically safety interlocked. The safety interlock circuit must not be bypassed or overridden.

ROUTINE CLEANING PROCEDURE	
Daily	Thoroughly clean the machine to remove any build up of dust, glue and product
	Use a soft brush and clean to remove dust, carton fragments etc.
	Wipe over loading table to remove any product and debris
	Wipe over guards with stainless steel cleaner and soft cloth
Glue Units	Wipe off excess glue from guides etc. Hint! Spraying areas around glue guns with silicone releasing agent before running will aid removal of any excess glue deposits.

e) Liability

You are not allowed to make any changes to the machine without the permission of T. Freemantle Ltd. This includes

- ◇ rebuilding machines, mounting and dismounting parts supplied by other manufacturers.
- ◇ rendering ineffective or removing safety devices.

Any unauthorized modification made on the machine will invalidate the manufacturers liability.

Failure to observe the safety regulations may:

- ◇ cause danger to the health of the user or to a third party
- ◇ cause damage to the equipment or other equipment
- ◇ impair the machine's efficiency

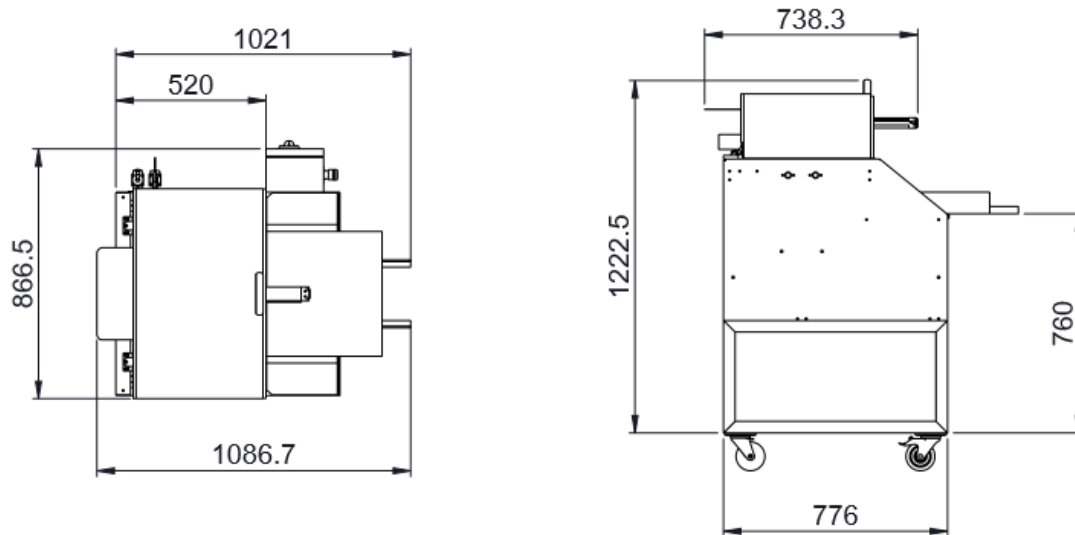
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B

SECTION 1

TECHNICAL DATA

1. B.1 Dimensions



1. B.2 Output

Single carton elevation up to 35 per minute.
 Double carton elevation up to 70 per minute.
 (Subject to size and quality of cartons)

1. B.3 Format Range

N.B. Restrictions apply to certain combinations of sizes.

Between Flaps-	50/300mm- min/max
Width	50/280mm- min/max
Height	20/120mm- min/max

1. B.4 Machine Data / Requirements

Compressed Air	80 psi / 5.5 bar
Electrical Power Supply	Standard 240V, 50Hz, Single Phase
Total Electrical Load	1 kW - Standard Machine

1. B.5 Sound Emission Data

The following illustration shows the decibel noise emission reading from the machine while it is in normal continuous production running.

The areas of probable operator input are highlighted on the drawing below. As the machine is a semi-automatic unit there is one defined workstation.

The readings are taken from a point 1 metre from the surface of the machine and 1.6 metres from the floor and are measured in decibels (dB).

(See Fig. 01)

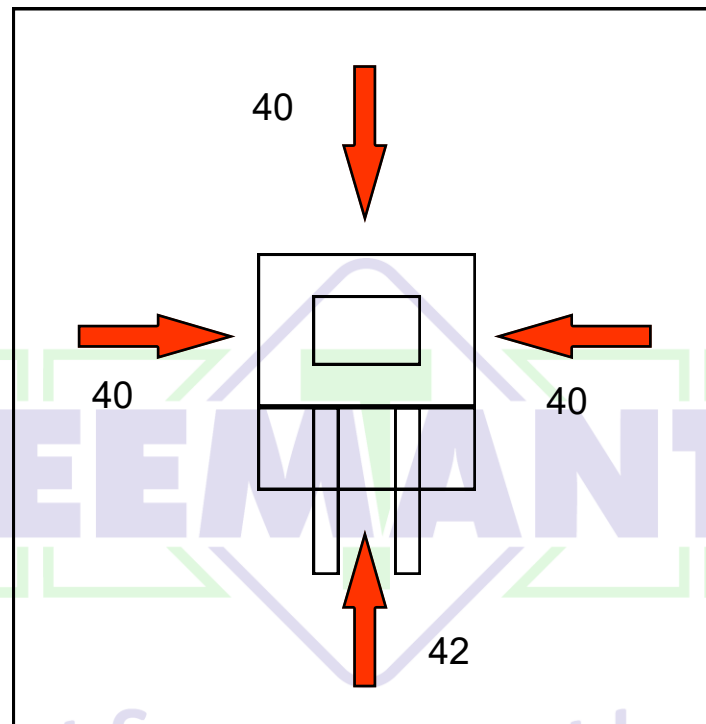


Fig.1

Note! Any modifications to the pneumatic circuit or the guarding system may affect sound emissions. Please refer to 1.A.3.e

SECTION 1

C DESCRIPTION OF MACHINE

1. C.1 General Description

a) Machine Construction

Construction	Rigid Steel one piece frame with Stainless Steel and Aluminium Construction. Comprising of Stainless Steel Side Plates and Shafts, Aluminium Tie Bars and Guides.
Mobility	Fully mobile on Lockable Castors (Optional Adjustable Jacking Feet).
Guarding	Brush Finish Stainless Steel Guarding, with Polycarbonate / Acrylic viewing stations.

b) Machine Specification

Product Infeed (Standard Machine)	Static Bed Product Infeed. 700mm Standard Machine.
Safety Interlock	Full Safety Interlock Guard System
Glue Applicator	Proven hot melt glue dabber system
Pneumatic System	ISO Standard maintenance free pneumatic system

c) Machine Options

Mounting Frame	Stainless Steel on lockable wheels
Glue Tank Options	Easy Fill Extended Volume
Glue system	Dual Glue Line Option
Product Infeed	Extended Infeed 1.5 metres Intermittent powered infeed
Carton Eject (when fitted)	Automatic inline system
Carton Coding (when fitted)	Contact Ink Coder End flap Deboss Coder

1. C.2 Machine Identification



Fig. 2

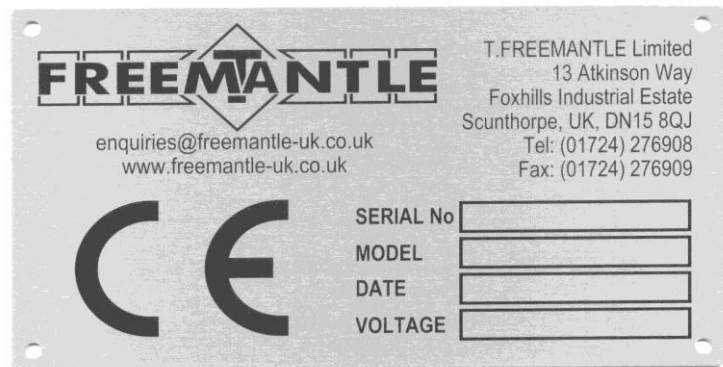


Fig. 3

The CE identification plate is located on the electrical panel, which is located on the right hand side of the machine. Always quote the information on this plate when making technical enquiries.

1. C.3 Machine Functions

Pre-loaded cartons are placed onto the product infeed. The cartons are then manually pushed towards the machine. The operator exerting forward pressure on the cartons activates the pneumatic cycle. The following operations can then be seen

1. Glue dabber cylinders extend and return.
2. Time delay.
3. Elevator extends and returns.
4. Push Off extends and returns (if fitted)

The sealed cartons are stacked in the “Chimney” until removed by the operator.

1. C.3. a) Product Infeed

The pre-loaded and folded cartons are placed onto the guide rails. The operator pushes the cartons and thus activates the pneumatic sequence. (Fig. 4)



Fig. 4

1. C.3. b) Carton Closure

The carton enters the gluing unit with both bottom flaps folded and top flaps in aeroplane format. (Fig. 5)

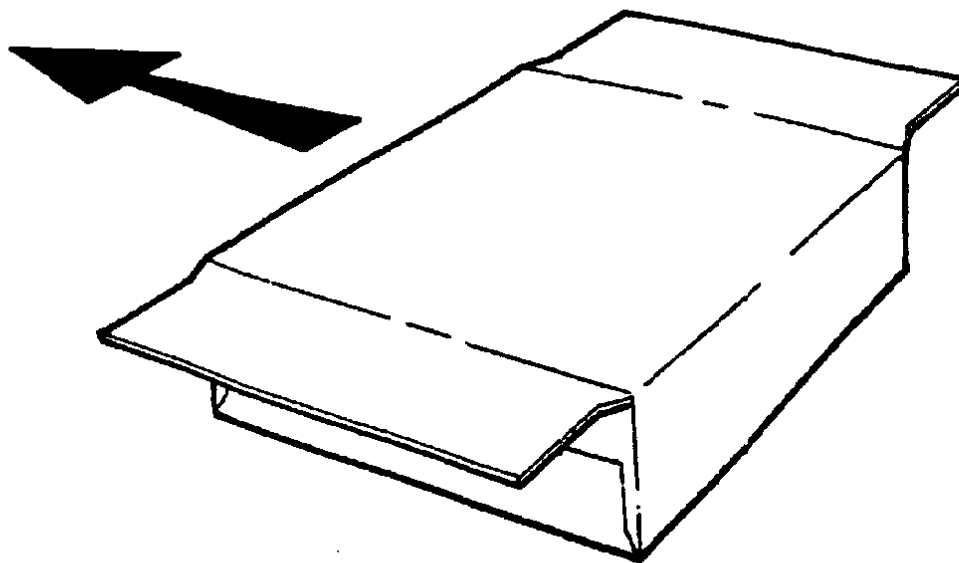


Fig. 5

Glue Line

The position of the glue line is controlled by sliding the tank assembly in or out depending upon carton size. (Fig. 6)

The glue dabbers should be applying glue to the underside of the major flap so as when the major flap is ploughed down the glue line is approx, 6mm from the edge. When the flap is ploughed down the glue line is compressed to form a good seal. The glue should give a good enough seal so that when the carton is opened the seal should tear the fibres of the carton board.



Fig. 6

**WARNING! THE GLUE UNIT WILL BE HOT DO NOT TOUCH
WITHOUT ADEQUATE PROTECTIVE CLOTHING**

1. C.3. c) Carton Discharge

After the glue has been applied, the top flaps are automatically ploughed down and the carton is then elevated out of the machine via the elevator cylinder. (Fig. 7)



Fig. 7

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SECTION 1

D **INSTALLATION AND CONNECTION**

Transportation

This machine has been designed and manufactured to give the machine excellent stability and maximum strength. For transportation purposes the machines are generally shipped in one piece. In special circumstances the machines may be shipped with guarding and control pod / panels removed to avoid damage. In the event of the machine needing any components to be removed for shipment then machinery is supplied with information outlining any re-assembly instruction that may be needed.

Installation

The machine have a guarantee of 12 month from date of installation, this guarantee is only applicable if the machinery is installed by a qualified T. Freemantle engineer.

Allocation should be made for connection to the correct power and compressed air services. (See Machine Data Section 1.B.4)

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SECTION 2

A INITIAL OPERATION

2.A.1 Preparing Machine for Operation

a) PRE-PRODUCTION CHECK LIST

Electrical Power	Turn Mains Isolator to 'ON' position (Fig. 8)
------------------	---



Fig.8

Glue Unit	The glue unit should be switched on at least 45 min's prior to production to allow the glue to reach its correct operating temperature. (FAILURE TO DO THIS COULD RESULT IN DAMAGE TO MACHINERY)
Hot Melt Glue Tank	Check that it is topped up. Note! If the glue level is allowed to drop too low then the introduction of too much fresh (cold) adhesive may result in a bad carton seal. Add fresh adhesive little and often.

Warning! The Glue Tanks must not be contaminated as this can result in a bad carton seal

Spraying areas around glue guns with silicone releasing agent before running will aid removal of any excess glue deposits.

a) PRE-PRODUCTION CHECK LIST CONTINUED

Machine Air Supply	Check that the Air Supply is set to the correct level on the Air Regulator Gauge.
Guarding	Check that these are closed.

2. A.2 Operator Input

Product infeed	Loading of the product into the carton is undertaken manually, the carton is then placed on the infeed guides.
Glue unit	During normal production running the glue units will periodically require refilling.
Product discharge	Product is normally taken from the machine manually and moved on through metal detectors, check weighers etc.

2. A.3 Housekeeping

Routine Cleaning Procedure

Daily	Thoroughly clean the machine to remove any build up of dust, glue and product
	Use a soft brush and clean to remove dust, carton fragments etc.
	Wipe over loading table to remove any product and debris
	Wipe over guards with stainless steel cleaner and soft cloth
Glue Units	Wipe off excess glue from guides and glue guns etc. Hint! Spraying areas around glue guns with silicone releasing agent before running will aid removal of any excess glue deposits.

Cartons

Carton manufacturers advise that cartons have a shelf life of around three months, providing they are stored away from moisture at temperatures of around 10 / 20 degrees C. If they are stored at varying temperatures or are allowed to become damp, then the shelf life will be reduced considerably, affecting both glue and all creases and therefore seriously reducing the production efficiency of the machine.

When the cartons are stored in cases attention must be given to the number of cases on a stack. If too many cases are stacked together then this may cause the bottom cases to be crushed.

Bent, twisted or damp cartons will cause problems during machine operation.

Adhesive

When any adhesive is used in a factory environment, contamination of that adhesive, in its raw cold state or in its molten state, it is almost inevitable unless strict housekeeping rules apply. Dirt in the gluing system will give rise to bad carton seals, resulting from incorrect glue application.

To avoid contamination keep the glue in its original container completely covered. When filling the glue tanks do not use cartons or such as scoops for transporting the adhesive from the container to the tank as dust or fragments of cardboard could fall into the tank. Dropped adhesive should be swept up and thrown away, do not return it to the container as this will inevitably contain dust and other debris.

Different grades of adhesive should never be mixed and strict control should be maintained on which adhesive is to be used on any specific line or carton.

WARNING! HOT MELT ADHESIVE WILL CAUSE SEVERE BURNS DO NOT ATTEMPT TO CARRY OUT ANY WORK ON OR AROUND THE HOT MELT UNIT WITHOUT ADEQUATE PROTECTIVE CLOTHING

SECTION 2

B **TROUBLESHOOTING**

2. B.1 **Machine Will Not Run**

Power Failure	Check the mains isolator is 'ON'. Check the circuit breakers and main fuses.
Insufficient Air Pressure	Check Air Pressure Regulator Gauge.
Safety Interlock Circuit	Check all guards are closed.

2. B.2 **Cartons Incorrectly Sealed**

Insufficient / no glue	Check glue tank level. Check glue dabbers for incorrect setting. Check machine width is set correctly to carton height. Check glue is up to operating temperature. Check that glue line is in unvarnished surface of board.
Glue Line in Wrong Position	Check Glue Dabbers positions and adjust if necessary.
Carton Flaps Weak Sealed	Check that the adhesive has the correct open time. Check that the amount of glue is not too small or too light. Check that there is sufficient compression on the glue line. Check that the adhesive temperature is not set too low or too high.

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SECTION 3

A MAINTENANCE SCHEDULE

**WARNING! SWITCH OFF ALL ELECTRICAL POWER BEFORE
CARRYING OUT ANY ELECTRICAL MAINTENANCE**

**N.B. ALL ELECTRICAL SWITCHES AND RELAYS HAVE A LIFETIME
OPERATION OF 500,000 OPERATIONS AND SHOULD BE CHECKED
PERIODICALLY**

**WE RECOMMEND THAT OUR SERVICE ENGINEERS COMPLETE AN
ANNUAL SERVICE. FOR FURTHER DETAILS PLEASE CONTACT OUR
SERVICE DEPARTMENT**

3. A.1. Maintenance Schedule

Daily	Clean machine thoroughly by removing build up of dust, glue or product. It is essential that this schedule is maintained to ensure that the machine continues to run smoothly. (See Routine Cleaning Schedule Section 2.A.3) Check the water separator on the air pressure regulator unit and drain the unit accordingly. Check air pressure and maintain the compressors under their own schedule.
Weekly	Check machine thoroughly for efficient operation and security of components. Check vacuum pipework for leaks.
Monthly	Check pneumatic system for leaks. Paying attention to the pipework directly under the glue tanks. Note! Ensure that any guarding that has been removed is replaced before the machine is put back into production.
Annually	Thoroughly check all components, guides and slides for signs of wear.

3. A.2. Maintenance of the Air Pressure Regulator

CHECK	Pressure gauge to ensure that the required operating pressure is correct
CHECK	Pneumatic hoses, unions and connections are all secure and free from leaks

3. A.3. Maintenance for the Electrical System

WARNING! SWITCH OFF ELECTRICAL POWER BEFORE CARRYING OUT ANY FORM OF MAINTENANCE

N.B. WE RECOMMEND THAT THE ELECTRICAL SYSTEM SHOULD BE CHECKED ON A REGULAR BASIS BY A QUALIFIED ELECTRICIAN

CHECK	That wiring and all electrically operated components are secure, undamaged and correctly enclosed Remake any damaged or defective wiring connections and replace any damaged components
CHECK	Thermostat for defects.
CHECK	Correct operation of safety guard interlock

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SECTION 3

B SIZE CHANGE

3. B.1. Change Procedures

Elevator Plate	The elevator plate is the only change part on the machine and is manufactured for specific carton sizes. To remove undo the countersink screws and change out the plate (Fig. 9)
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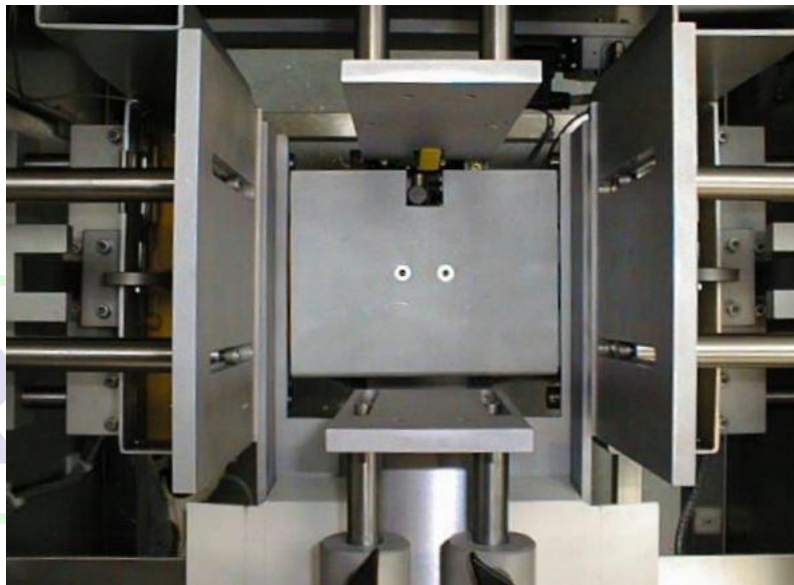


Fig. 9

For Height of Carton	Adjustment is obtained by undoing the cap heads on the two side plates and the front plate, it is advisable to utilise a carton to achieve the correct height. (Fig. 10/11)
----------------------	---



Fig. 10



Fig. 11

3. B.1. Change Procedures Continued

For Width of Carton	Adjustment is obtained by unlocking the handles on the front and back plates. As a guide the carton should be able to slide freely in the plates. Ensure the plates are positioned centrally over the elevator. (Fig. 12)
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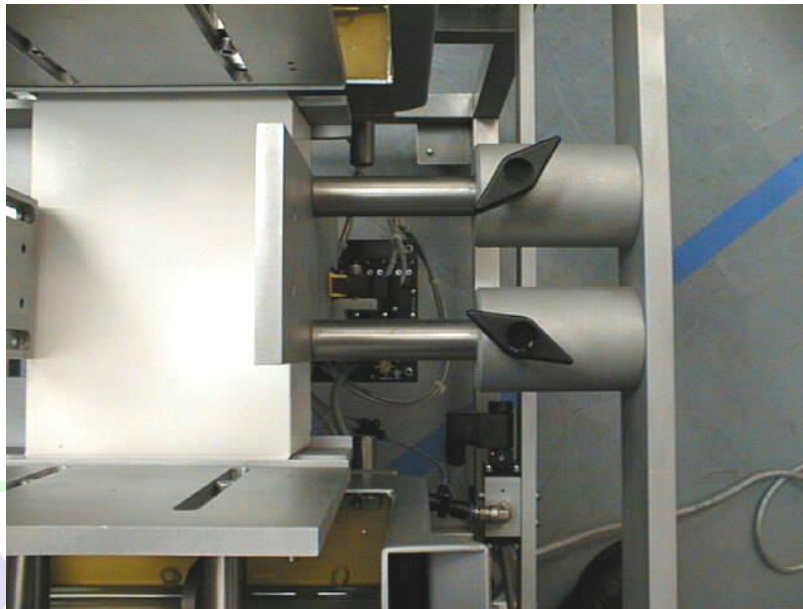


Fig. 12

For Between Flaps	Adjustment is obtained by unlocking the handles on the side plates. As a guide the carton should be able to slide freely in the plates. Ensure the plates are positioned centrally over the elevator. (Fig. 13)
-------------------	---



Fig. 13

3. B.3 Carton Gluing Change

Glue Unit Position	Adjustment of the glue dabber position is achieved by loosening the locking handle and moving the tank in or out to the required position As a guide the glue dabbers should be applying glue to the flap so as when the major flap is ploughed down the glue line is approx. 6mm from the edge.(Fig. 14)
--------------------	--



Fig. 14

Gluing Height Restriction	To facilitate changes in carton height the dabbers are restricted. To adjust undo the locking screws and slide the stop to the desired position (Fig. 15)
---------------------------	---



Fig. 15

WARNING! THE GLUE UNIT WILL BE HOT DO NOT TOUCH WITHOUT ADEQUATE PROTECTIVE CLOTHING

A

SECTION 4

HOT MELT ADHESIVE

Introduction

In their solid state these products are generally free of hazard. However, in their molten state there are potential hazards whose significance depends on the individual situation, particularly since the amount of product involved varies widely from one application to another.

Hazards

Burns

Burns are undoubtedly the major source of injury and they could be severe since application temperatures are usually in the range 120°C-240°C. They may result from either contact with the product itself or from contact with equipment used for handling or applying it.

Fumes

Under normal operating conditions in well-ventilated areas, fumes do not constitute a serious hazard. They are, however, composed of low molecular weight organic materials which are unpleasant and which may be harmful or irritating to the respiratory system. If the vapour concentration is high enough, they may constitute an explosion or fire hazard.

Hot melt products degrade to varying extents from the effect of heat and oxygen and the effects are directly related to time, temperature and formulation. When grossly overheated (at least 30°C-40°C above the recommended operating temperature), the effects can be severe and some of the decomposition products could pose a serious risk to health.

Fire

There is no particular fire hazard with hot melts, but in a fire situation they are combustible and will burn with the evolution of black acrid smoke. It should be noted that a degraded hot melt generally has a lower auto-ignition temperature than the fresh product.

Essential Safety Precautions

Burns

Avoid splashing and take special care when stirring or adding product to a reservoir. Do not allow molten product to come into contact with the skin or eyes. Face and eye protectors conforming to BS.2092 must be worn when there is any possibility of splashing occurring.

Protective clothing must be kept in good condition and worn at all times when handling molten adhesive or hot equipment.

Fumes

DO NOT breathe fumes.

Adequate extraction must be provided to ensure exposure is kept as low as is reasonably practicable. See HSE Guidance Notes EH18, EH22 and EH40 (revised annually).

Cleanliness: the equipment should be kept clean and free from build-up of solidified material which on degradation could possibly form harmful or irritating vapours.

It is the users responsibility that the hot melt adhesive doesn't contain ingredients that will pose a risk.

Fire

Ventilation ducts must be inspected regularly and kept clean to prevent the build-up of condensate which could produce a fire or explosion risk.

Storage

Every precaution must be taken to avoid hot melt adhesives or sealant becoming moist from condensation or other sources since adding wet material to already molten products is likely to cause spitting.

Emergency Actions**Burns**

Immediately treat the affected area by immersion or irrigation with clean cold water until the adhesive or sealant is cold.

DO NOT ATTEMPT TO REMOVE THE ADHESIVE OR SEALANT FROM THE SKIN.

Cover the affected area with a cold compress or dry sterile gauze dressing.

Obtain medical attention immediately.

Fumes

If exposure does result in discomfort, the affected person should be removed to fresh air, kept warm and at rest. Medical advice should be sought.

Fires

Switch off source of heat.

Small fires may be tackled with dry powder or carbon dioxide extinguishers.

DO NOT USE WATER.

HEATER

Heater Type: GOTHERM FP 0269A

Specification: 240v 500W

